

ME 333
MECHANICS OF MATERIALS

Lecture 15: Bending of Beams II - Review

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Sem I, 2025 - 2026

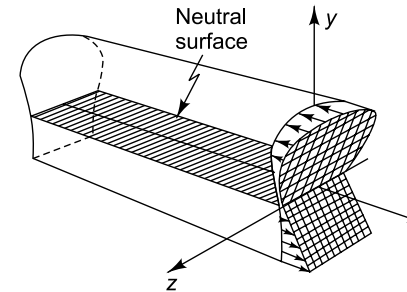
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Pure Bending

Static Equilibrium Conditions



- In case of pure bending, the resultant normal forces acting on the cross-sectional area must be zero.

$$\int_A \sigma_x dA \Rightarrow \int_A \left(\frac{-y}{\rho} E \right) dA = 0$$

$$\Rightarrow \int_A y dA = 0$$

- The neutral axis passes through the centroid of the cross-section (as long as stresses are in the elastic range)

Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

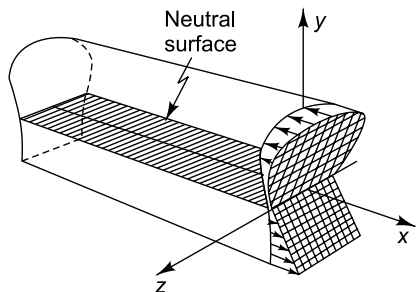
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Pure Bending

Static Equilibrium Conditions



- In case of pure bending, the moment of internal forces about neutral axis must be equal to M.

$$\int_A (-y) \sigma_x dA = \int_A (-y) \left(\frac{-y}{\rho} E \right) dA = M$$

$$\frac{E}{\rho} \int_A y^2 dA = \frac{E}{\rho} I_{zz} = M \Rightarrow \frac{1}{\rho} = \frac{M}{EI_{zz}}$$

$$\therefore \sigma_x = \frac{-y}{\rho} E = -\frac{My}{I_{zz}}$$

Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

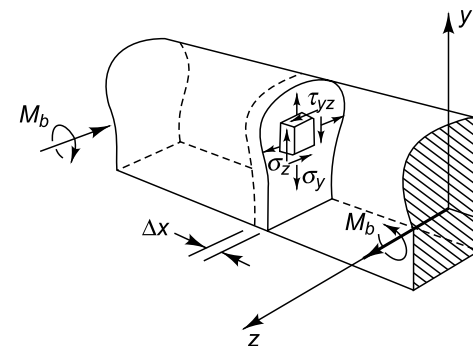
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Pure Bending

Stresses in pure bending



Assumption :

$$\sigma_y = 0$$

$$\sigma_z = 0$$

$$\tau_{yz} = 0$$

Slenderness of the beam allows for these assumptions

Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

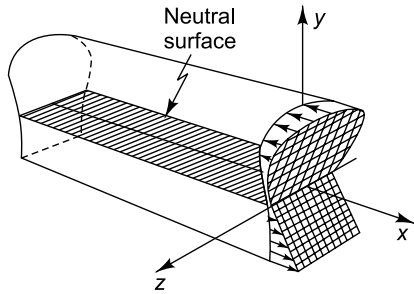
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Pure Bending

Neutral Axis



Inferences :

- the first moment of cross-sectional area about the neutral axis must be zero
- It implies that the *neutral axis must pass through the centroid of the cross-section.*
- An extension of above inferences is that the neutral surface must pass through the centroid of the cross-sectional area

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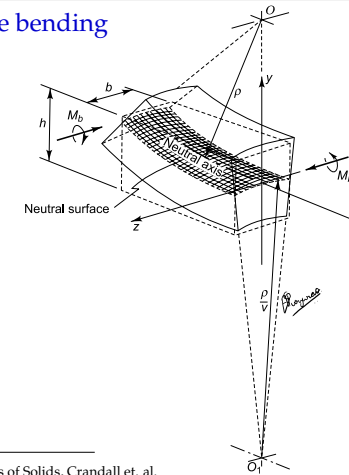
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Pure Bending

Deformed Beam under pure bending



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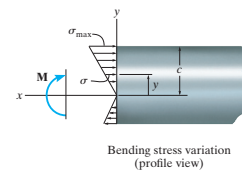
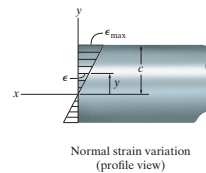
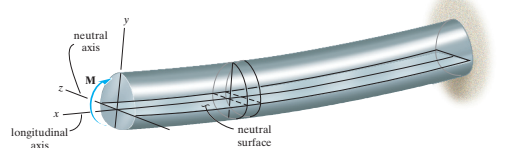
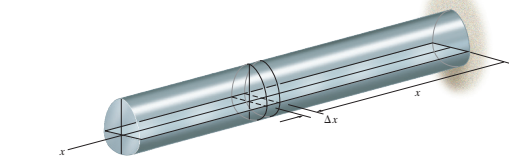
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Pure Bending

Flexure Formula : $\sigma_x = -\frac{My}{I_{zz}}; \quad \epsilon_x = -\frac{My}{EI_{zz}}$



Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

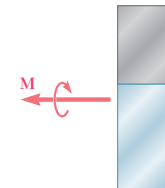
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Composite Beams



- Consider a beam cross-section made of two different materials that are perfectly bonded.
- Normal strain still varies linearly with distance y from the neutral axis of the cross-section

$$\epsilon_x = -\frac{y}{\rho}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

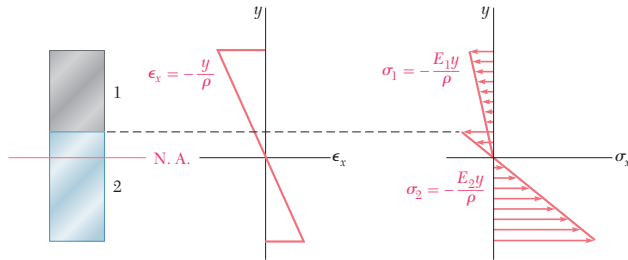
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Composite Beams



- No assumption was made in the derivation of the relationship regarding the stress-strain response of material.
- There are jumps in stresses at the material interface.

Q Where is the Neutral axis ?

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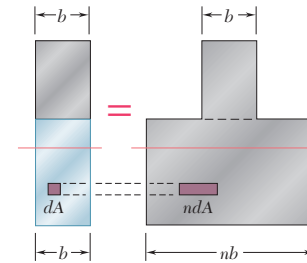
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Composite Beams

Method of transformed sections



Ref Base Fig: Mechanics of Materials, Beer & Johnston

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- N.A comes from normal force equilibrium :

$$\sum F_x = 0 \implies \int_A \sigma_x dA = 0$$

- Normal stress in the two materials is:

$$\sigma_x = \begin{cases} -E_1 y / \rho & \text{Mat 1} \\ -E_2 y / \rho & \text{Mat 2} \end{cases}$$

- Incremental force measures:

$$\begin{aligned} dF_1 &= \sigma_1 dA = -(E_1 y / \rho) dA \\ dF_2 &= \sigma_2 dA = -(E_2 y / \rho) dA \\ &= -\frac{n E_1 y}{\rho} dA = -\frac{E_1 y}{\rho} (n dA) \end{aligned}$$

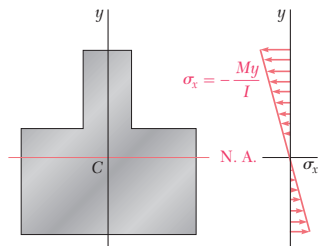
Ref Base Fig: Mechanics of Materials, Beer & Johnston

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Composite Beams

Method of transformed sections



- It means that the resistance to bending of the beam would remain same if both portions were made of the first material, provided that the width of each element of lower portion were multiplied by the factor n
- Note that in a direction parallel to NA :

$$\begin{aligned} n > 1 & \text{ means widening} \\ n < 1 & \text{ means narrowing} \end{aligned}$$

- In Composite beams, the centroidal axis does not coincide with the neutral axis.

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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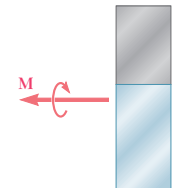
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Composite Beams

Analytical Method



- N.A comes from normal force equilibrium :

$$\begin{aligned} \sum F_x &= 0 \implies \int_A \sigma_x dA = 0 \\ \implies \int_{A_1} \sigma_{x1} dA + \int_{A_2} \sigma_{x2} dA &= 0 \\ \implies \int_{A_1} -\frac{E_1}{\rho} y dA + \int_{A_2} -\frac{E_2}{\rho} y dA &= 0 \\ \implies E_1 \int_{A_1} y dA + E_2 \int_{A_2} y dA &= 0 \end{aligned}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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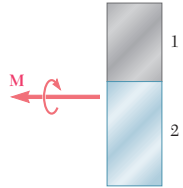
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Composite Beams

Analytical Method



- Moment curvature relationship :

$$\begin{aligned}\sum M_z &= - \int_A \sigma_x y dA = M \\ \Rightarrow - \int_{A1} \sigma_{x1} y dA - \int_{A2} \sigma_{x2} y dA &= M \\ \Rightarrow \int_{A1} \frac{E_1}{\rho} y^2 dA + \int_{A2} \frac{E_2}{\rho} y^2 dA &= M \\ \Rightarrow \frac{1}{\rho} (E_1 I_1 + E_2 I_2) &= M\end{aligned}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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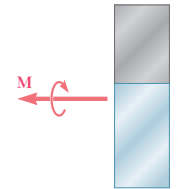
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Composite Beams

Analytical Method



- Strains in beam :

$$\epsilon_x = -\frac{y}{\rho} = -\frac{My}{(E_1 I_1 + E_2 I_2)}$$

- Stresses in beam :

$$\sigma_{x1} = E_1 \epsilon_x = -\frac{My E_1}{(E_1 I_1 + E_2 I_2)}$$

$$\sigma_{x2} = E_2 \epsilon_x = -\frac{My E_2}{(E_1 I_1 + E_2 I_2)}$$

- In Composite beams, the centroidal axis does not coincide with the neutral axis.

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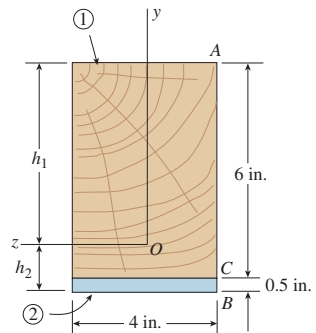
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Composite Beams

[Example 1] Determine the location of Neutral Axis, given the modular ratio $E_2/E_1 = 20$.



- Step 1 : From normal force equilibrium we know that:

$$\begin{aligned}E_1 \int_{A1} y dA + E_2 \int_{A2} y dA &= 0 \\ \Rightarrow (Q_z)_1 + 20(Q_z)_2 &= 0 \\ \Rightarrow A_1 \bar{y}_1 + 20 A_2 \bar{y}_2 &= 0 \\ \Rightarrow 24(h_1 - 3) + 20 \times 2(-h_2 + 0.25) &= 0 \\ \Rightarrow 3h_1 - 5h_2 &= 7.75\end{aligned}$$

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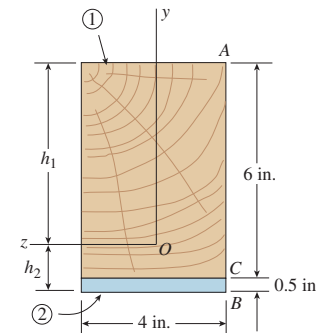
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Composite Beams

[Example 1] (Contd ...)



- Step 2 : Solving the system of equations simultaneously:

$$3h_1 - 5h_2 = 7.75$$

$$h_1 + h_2 = 6.5$$

We get: $h_1 = 5.031$ in and $h_2 = 1.469$ in

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Eccentric Loading

Combined axial and bending loads



Walkway light.



Bench press.

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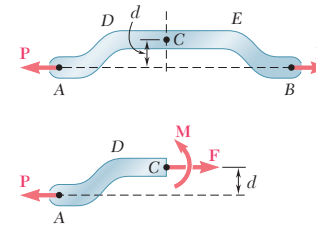
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Eccentric Loading

Combined axial and bending loads



- Occurs if we have non zero bending moment (M) and axial force (P) at a cross section
- Based on static equilibrium conditions :

$$\sum F_x = 0 \implies F = P$$

$$\sum M_c = 0 \implies -Pd + M = 0 \quad \therefore M = Pd$$

- Normal stress is a superposition:

$$\begin{aligned}\sigma_x &= \sigma_{x(\text{axial})} + \sigma_{x(\text{bending})} \\ &= \frac{P}{A} - \frac{My}{I}\end{aligned}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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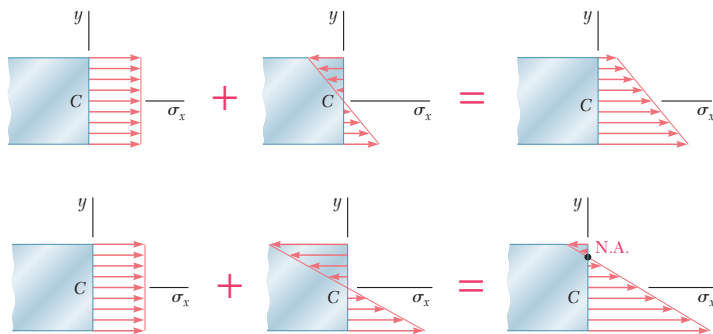
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Eccentric Loading

Combined axial and bending loads



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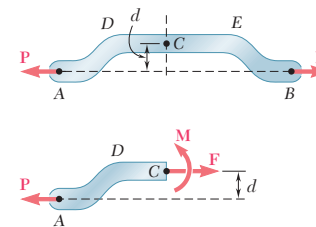
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Eccentric Loading

Combined axial and bending loads



- In case of combined loading, Neutral axis does not coincide with the centroidal axis of the section ($\sigma_x \neq 0$ for $y = 0$)
- The validity of the formula is restricted to scenarios that satisfy the following conditions :
 - Stress must not exceed proportional limit of loading
 - Deformations due to bending must not appreciably affect the distance d
 - Computation at section points must not be close to points near E

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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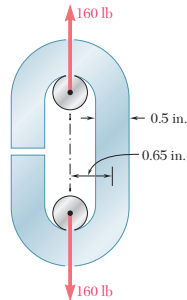
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Eccentric Loading

[Example 2]: An open-link chain is obtained by bending low-carbon steel rods of 0.5 in. diameter into the shape shown. Knowing that the chain carries a load of 160 lb, determine (a) the largest tensile and compressive stresses in the straight portion of a link, (b) the distance between the centroidal and the neutral axis of a cross section.



Ref Base Fig: Mechanics of Materials, Beer & Johnston

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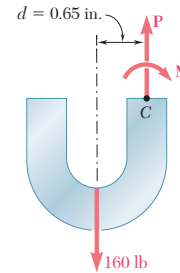
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Eccentric Loading

[Problem 1]: a) Largest tensile or compressive stress



- Determine geometric measures :

$$A = \pi \times 0.25^2 = 0.1963 \text{ in}^2$$

$$I = (\pi/4) \times 0.25^4 = 3.068 \times 10^{-3} \text{ in}^4$$

- Determine the bending moment :

$$M = Pd = 160 \times 0.65 = 104 \text{ lb-in}$$

- Determine the axial & bending stresses :

$$\sigma_x|_{\text{axial}} = \frac{P}{A} = \frac{160}{0.1963} = 815 \text{ psi}$$

$$\sigma_x|_{\text{bend}} = \pm \frac{Mc}{I} = \pm \frac{104 \times 0.25}{3.068 \times 10^{-3}} = 8475 \text{ psi}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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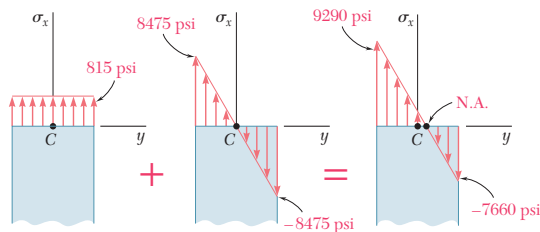
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Eccentric Loading

[Problem 1]: a) Largest tensile or compressive stress (Contd ...)



- Largest tensile and compressive stress is:

$$\sigma_t = \sigma_x|_{\text{axial}} + \sigma_x|_{\text{bend}} = 815 + 8475 = 9290 \text{ psi}$$

$$\sigma_c = \sigma_x|_{\text{axial}} - \sigma_x|_{\text{bend}} = 815 - 8475 = -7660 \text{ psi}$$

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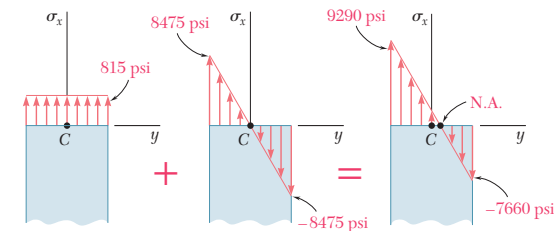
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Eccentric Loading

[Problem 1]: b) Distance between centroidal and neutral axis



- N.A. is located where $\sigma_x = 0$, i.e.,

$$\sigma_x = \frac{P}{A} - \frac{My}{I} = 0$$

$$\Rightarrow y = \frac{PI}{AM} = \frac{160 \times 3.068 \times 10^{-3}}{0.1963 \times 104} = 0.0240 \text{ in}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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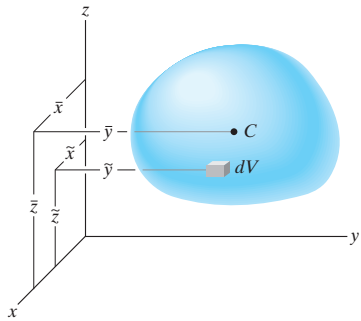
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Moments of Area

Centroid of Volume :



$$\bar{x} = \frac{\int_V \tilde{x} dV}{\int_V dV}$$

$$\bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV}$$

$$\bar{z} = \frac{\int_V \tilde{z} dV}{\int_V dV}$$

Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

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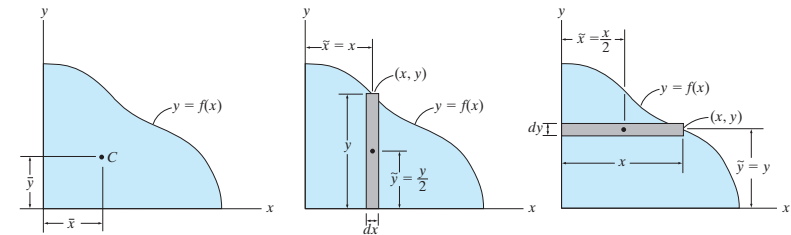
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Moments of Area

Centroid of an Area :



$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA};$$

$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

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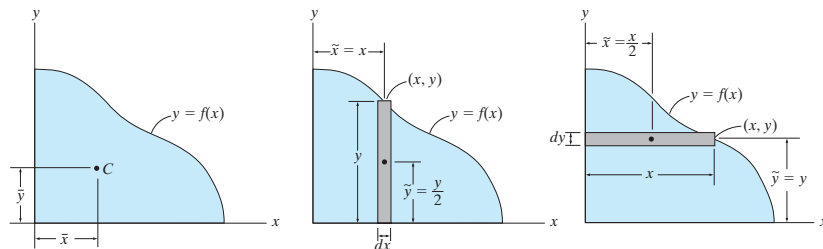
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Moments of Area

Centroid of an Area :



- Centroid coincides with center of gravity of a material that is uniform or homogeneous (constant density or specific weight)
- Centroid is the moment balance of sum of area moments of all points and the area moment of the resultant.

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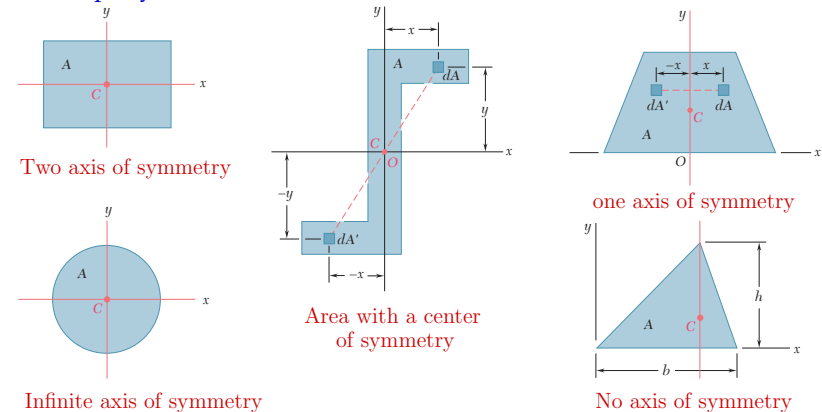
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Moments of Area

Centroid Property :



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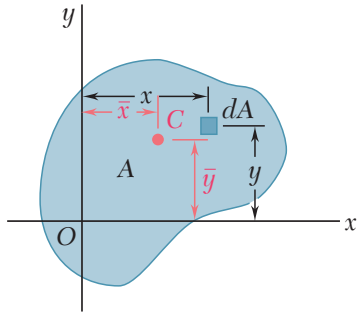
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Moments of Area

First Moment of Area :



$$\bar{x} = \frac{\int_A x dA}{\int_A dA} = \frac{Q_y}{A}; \quad \bar{y} = \frac{\int_A y dA}{\int_A dA} = \frac{Q_x}{A}$$

$$Q_y = A\bar{x}; \quad Q_x = A\bar{y}$$

- If Q is zero then the centroid coincides with the origin of the coordinate axis.
- The first moment of area about the axis of symmetry is zero.

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Reading Assignment

Chapter 7 Crandall et. al.

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